How a decision is made, tested and proven

Nine documents consolidated into one, with the eight places they disagreed printed rather than resolved. Decision points, strategies and flows; routing and policy; assignment, attribution and promotion; and the vocabularies that have to be settled before any of it can be authored.

Document	PD-002 · v1.0
Date	27 September 2026
Consolidates	9 documents, 165 sections
Companion	PD-001 · the platform
Unresolved conflicts	8, printed in section 01
Status	Canonical — supersedes its sources

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PART

I — The model

01 How to read this document

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How to read this document

> This consolidates nine documents that between them describe one system. They were written over several months by people reading each other's output, and in eight places they disagree.

SOURCE	WHAT IT CONTRIBUTED	STATUS
Decision platform PRD	The core model, the registry, the lifecycle, the promotion gates	Absorbed
Flow and data model	The one-idea framing, the record shapes, the failure modes	Absorbed
Worked example	Seven findings, four of them still open, and the assignment correction	Absorbed
Definitions and readiness	Posture, bindings, the vocabularies, the glossary, the readiness gates	Absorbed — the most cited part of the corpus
Routing strategy PRD and design	Contracts, objective forms, the maturity model, the moat	Absorbed
Routing strategy catalogue	Nine families of strategy	Absorbed **as a checklist**, not as a deployment plan
Routing strategy selection	The four layers and the simplification argument	Absorbed
Routing and decision model	The five decision points, the tuple, the segment profiles	Absorbed
Policy platform PRD and design	The algebra, namespaces, compilation, the policy shapes	Absorbed

THE EIGHT CONFLICTS, PRINTED RATHER THAN QUIETLY RESOLVED

Four have a clear right answer and are resolved here. Four do not, and are questions about what we want rather than about what was written. A consolidation exercise is the wrong place to settle those, so both readings are printed and each carries an owner.

#	SUBJECT	THE DISAGREEMENT	RESOLUTION
C1	**Where assignment happens**	PRD and flow model: at context assembly. Worked example: at the divergence point, explicitly correcting the PRD.	**Resolved** — the divergence point. Section 04 shows the arithmetic. The PRD amendment is still open and blocks full ramp.
C2	**How many decision points exist**	Five in the PRD. The flow model adds message augmentation as a sixth. The worked example uses eight identifiers.	**Resolved** — five in the registry; message augmentation is a registration, not a sixth in principle. Section 03.
C3	**Unit of randomisation for credential selection**	PRD: the credential. Worked example: the customer, arguing the PRD is wrong because randomising one member of a set does not define an arm.	**Resolved** — the customer. The argument is correct.
C4	**The name of the balanced routing binding**	`Route_Balanced` in three documents, `Route_BalancedEconomics` in the catalogue.	**Resolved** — `Route_Balanced`.
C5	**The treatment fidelity floor, second clause**	Below 95%, ***or three points below control***. Versus: below 95%, ***or a gap of more than two points between arms***.	**Open.** Both printed in section 14. Owner: data science. These are not the same threshold at low volume.
C6	**Fraud and dispute guardrail units**	+15% and +10% relative. Versus 1.5× and 1.3× control at a minimum sample.	**Open.** Both printed. Owner: risk.
C7	**The strategy catalogue count**	Fifty-two across nine families, and the derived "other forty-seven". The tables enumerate seventy-three.	**Resolved** — seventy-three. The simplification argument is unaffected; the derived figure is dropped.
C8	**The identifiers in the worked example**	The two sub-decision-points are labelled inconsistently between the current-state and new-flow notes.	**Open, and blocking.** They land in strategy names, layers, decision records and every readout. Owner: architecture. Freeze before registration.

Decision point, strategy, flow

THE ONE IDEA · EVERYTHING ELSE IS DOWNSTREAM OF IT

A VARIANT FLOW DIFFERS FROM ITS BASELINE AT EXACTLY ONE DECISION POINT

diff_count = 1, enforced by the compiler at registration. Not a guideline.

DECISION POINT

a place in the flow where a choice is made

input schema · action schema · terminal fallback · latency budget · experimentable

STRATEGY

how one decision point decides for one population

class · population · objective · fallback · exploration policy · expiry

FLOW

one strategy bound at every decision point, no gaps

role · baseline_ref · bindings · diff_count · signature · ttl

AN EXPERIMENT SPLITS TRAFFIC BETWEEN A BASELINE FLOW AND A VARIANT

Arms sum to 1.0 less the permanent holdout. Three arms beat two experiments — more power, and no contention.

— If a variant changed two things and won by 37 bps, nothing in the data says whether the first change produced 37, the second produced 37, or the first produced 55 and the second cost 18. That is the whole reason for the rule.

OBJECT	DEFINITION	FIELDS THAT DO THE WORK
Decision point	A place in the flow where a choice is made.	`input_schema` (the context subset it may read) · `action_schema` · `constraint_binding` · `default_strategy` · `terminal_fallback` (deterministic, with no fallback of its own) · `latency_budget_ms` (hard, enforced by the orchestrator) · `experimentable` · `experiment_layer` with its own salt
Strategy	How one decision point decides for one population. Chooses between options that are all permitted and all possible.	`class` (deterministic, parametric, learned) · `population` (must not intersect another binding at this point) · `fallback_strategy_id` (**required, never implicit**) · `objective_ref` · `exploration_policy` in basis points · `required_features` with minimum freshness · `minimum_evidence` · `attribution_baseline` · `expiry` (mandatory)
Flow	A complete, named, versioned way of processing one transaction type, with one strategy bound at every decision point.	`role` (baseline or variant) · `baseline_ref` · `bindings[]` with no gaps · `diff_count` · `differs_at` · `signature` · `ttl`
Experiment	Splits traffic between a baseline flow and one or more variants.	`hypothesis` (one sentence, written before launch) · `primary_metric` (**exactly one**) · `guardrails[]` · `unit_of_randomisation` · `sample_size` pre-registered with the effect size it detects · `stopping_rule` · `hard_expiry`

THE RULE EVERY OTHER RULE IS DOWNSTREAM OF

`diff_count = 1`. A variant flow may differ from its baseline at exactly one decision point, enforced by the compiler at registration.

The reason is not tidiness. If a variant changes two things and wins by 37 basis points, nothing in the data says whether the first change produced 37, the second produced 37, or the first produced 55 and the second cost 18. An experiment that cannot answer its own question at any confidence has consumed traffic and produced nothing.

The natural way to break it: both changes were ready at the same time. That is the most common registration rejection, and it is why the gate is mechanical rather than a review conversation. [A]

Three arms are preferred to two experiments. Two ideas at the same decision point become one experiment with three arms and a shared control: more power per unit of traffic, and it is the only form that does not contend for the layer.

What the compiler enforces, and what it rejects

RULE	REJECTS
`diff_count = 1` on a variant	Two changes in one flow
Binding completeness	A flow with any unbound decision point
Experimentability	A variant differing at a non-experimentable decision point
Population containment	A bound strategy whose population does not contain the transaction type
Signature	An unsigned flow is never served
Fallback chain	An implicit fallback, or a chain that does not terminate. **Depth is capped at two**: strategy → fallback → terminal
Naming	Sequential names. `MsgAug_DropCVV_v1` passes; `Route_Strategy1` does not — the name appears in the readout, the ticket and the incident
Expiry	Anything without a review date and an expiry
Pre-assignment identity	Arms whose graphs differ before the assignment node, verified by diffing them

The decision point registry

> Five at launch, extensible. The platform is decision-point agnostic — adding a sixth is a registration, not a change to the platform.

ORDER	DECISION POINT	CHOOSES	WHEN IT IS NOT A CHOICE	LAYER SALT
1	**Rail selection**	Card, ACH, RTP, FedNow, wire, open banking, wallet	Most consumer ecommerce offers one rail, so the capability filter decides	salt_5
2	**Credential selection**	Network token, FPAN, acquirer token, alternate stored credential	When only one credential form is eligible	salt_4
3	**Authentication**	None, exemption, 3DS frictionless, challenge, delegated, passkey	When SCA rules force it, or the market has none	salt_1
4	**Routing**	Processor, acquirer, network, message model, merchant account	When capability and policy leave one candidate	salt_2
5	**Attempt strategy**	What to try next, and when	On the first attempt	salt_3

Points two and four are coupled, which is what motivates the candidate tuple in section 06. Rail selection being a real choice is a correction of an earlier model that collapsed rails into the capability filter.

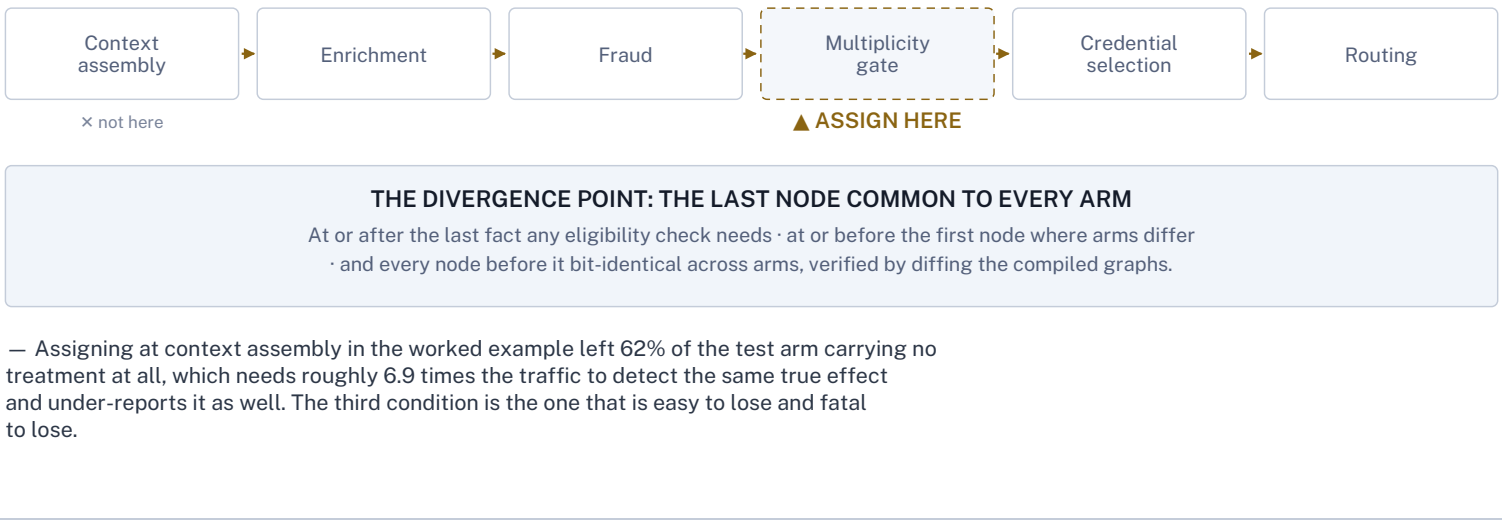
WHAT IS NEVER EXPERIMENTABLE, AND THE DISTINCTION THAT MATTERS

Screening. Constraint enforcement. Anything altering the amount charged, the currency, or a disclosure flow. Emergency propagation.

The distinction people lose: **the screening result remains available as a feature to other decisions. It is the decision that is not experimentable, not the data.** Risk is likewise not experimentable here, because risk owns its own promotion path — its output is an input to these decisions, not one of them.

Each experimentable decision point is a ***layer*** with its own hash salt, so assignment across layers is independent. Exactly one active experiment is permitted per layer per segment, and the allocation ledger enforces that against a controlled segment vocabulary — which does not yet exist, and is open item D17.

Assignment



— Assigning at context assembly in the worked example left 62% of the test arm carrying no treatment at all, which needs roughly 6.9 times the traffic to detect the same true effect and under-reports it as well. The third condition is the one that is easy to lose and fatal to lose.

Mechanics are deliberately dull: `bucket = hash(unit_id || salt_layer) mod 10000`, computed once, carried on the context, never recomputed. A retry of the same transaction lands in the same arm. Propensity is recorded as the arm's allocation share.

The three conditions an assignment point must satisfy

#	CONDITION	WHY
1	At or after the last fact any eligibility check depends on	Otherwise eligibility is evaluated against information that has not arrived.
2	At or before the divergence point — the first node where arms differ	Otherwise the arms have already diverged before anyone was assigned to them.
3	**Every node before the assignment point is bit-identical across all arms**	This is the one that is easy to lose and fatal to lose: if it fails, the assigned population has already been shaped by the treatment. The compiler verifies it by diffing the graphs up to the assignment node.

WHAT ASSIGNING TOO EARLY COSTS [C]

In the worked example, of 10,000 transactions 6,200 are single-credential and never reach the decision point being tested. Assigned at context assembly, **62% of the test arm carries no treatment at all** — assigned but untreatable.

The consequence is quantified: the experiment needs roughly **$1 / 0.38^2 \approx 6.9$ times** the traffic to detect the same true effect, and it under-reports that effect as well. Both directions are bad and the second is worse, because it is invisible.

This is the correction that gives PD-002 its assignment rule, and amending the source PRD is still an open item that blocks full ramp.

Eligibility — eight checks, any false short-circuits to baseline

CHECK	REASON CODE WHEN IT FAILS
Population predicate	`POPULATION_MISS`
Treatment availability	`MISSING_TREATMENT_CREDENTIAL`
Merchant opt-out	`MERCHANT_OPTED_OUT`
Strategic account	`STRATEGIC_ACCOUNT`
Permanent holdout	`HOLDOUT`
Migration soak slice	`MIGRATION_SOAK` — migration wins; experiments wait
Layer occupancy	`LAYER_OCCUPIED`
Autonomy floor	`AUTONOMY_BELOW_REQUIRED`

The second is the one that gets written wrong. **Eligibility must require everything the treatment needs, not just everything the hypothesis mentions.** A predicate that selects customers with more than one credential, without requiring the specific credential the treatment substitutes to, assigns transactions that then fall back — diluting the effect and dragging fidelity down.

Units of randomisation

DECISION POINT	UNIT	WHY NOT LOWER
Attempt strategy	Transaction	Later attempts are caused by earlier ones
Authentication	Transaction	The authentication context binds to the transaction and survives retries
Routing	Transaction, or customer where issuer-side learning is plausible	Issuer behaviour adapts to repeated exposure from one cardholder
Credential selection	**Customer**	The credential set belongs to the customer. Randomising on one member does not define an arm — and flipping a stored credential's form between transactions is itself a risk signal to the issuer, so the arms contaminate each other through the issuer's own model
Rail selection	Transaction	—
Any pricing-affecting change	Merchant	Charging two customers of one merchant differently is a contractual problem, not a statistical one

A unit that cannot be resolved — a guest checkout with no customer reference, when the unit is the customer — is excluded as a population miss rather than assigned on a fallback key. **A mixed-unit design is worse than a smaller one.**

PART

II — Routing and policy

05 The four layers

06 Bindings, postures and objectives

07 Fourteen segment profiles

08 Recovery is not ordinary routing

09 The policy platform

10 Four policy shapes

The four layers

FOUR LAYERS · ONLY ONE OF THEM CHOOSES ANYTHING

1 CAPABILITY FILTER — can this route do it at all?
deterministic · no owner beyond the capability matrix

2 POLICY CONSTRAINTS — is it permitted?
authored, absorbing, audited

ACTION SPACE = what is left

3 ONE STRATEGY — which of the remaining?
selected by the merchant's posture — the only layer with a choice

4 FEATURES — what the strategy reads
no bindings, no contention, no attribution question

FIVE BINDINGS PLUS A TERMINAL, SERVING FOURTEEN SEGMENT PROFILES
MaxApproval · Balanced · MinCost · Directed · Recovery · and StaticPreference, which has no fallback of its own.

— The separating test: a strategy chooses between options that are all permitted and all possible. If the answer is forced, that is not a choice. Applying it eliminated roughly nine of every ten entries in the original catalogue.

> The catalogue enumerates seventy-three named routing strategies. Five of them are strategies. The other sixty-eight are capability predicates, policy, or features — and the test that sorts them is one sentence.

> **A strategy chooses between options that are all permitted and all possible. If the answer is forced — because the route physically cannot do what the transaction needs, or because a rule forbids it — that is not a choice.**

Applying that test eliminates roughly nine in ten entries. A catalogue of seventy-three named strategies is a good inventory of things to think about and a terrible operating portfolio: every one would need an owner, an expiry, a review, a guardrail set, attribution and a contention check.

Where the sixty-eight went

CATALOGUE ENTRIES	BECOMES	LAYER
Lodging, airline, fuel, restaurant tip, healthcare, marketplace split, unattended, B2B enhanced data, gig disbursement, education, insurance, utility, QSR low ticket, digital goods, gaming, subscription MIT, partial auth, incremental, account funding, instalment, network token aware	Capability predicates	1
Data residency, domestic scheme mandated, surcharge compliance, sanctions corridor avoidance, SCA awareness	Policy — compliance is never a preference	2
Diversification, concentration management	Policy, as an **economic constraint** — an optimiser that can trade concentration will concentrate	2
Health weighted, circuit breaker, latency aware, failover, capacity shaped, maintenance window aware, regional isolation	Features — health is an input to every strategy, not a competing strategy	4
Issuer affinity, MCC optimised, amount band tuned, time of day aware, credential age aware, token form aware, updater state aware, fresh credential aware	Features of the approval model	4
Scheme fee aware, FX cost aware, processor rate card, interchange qualification, cross-border fee avoidance, least cost debit, PINless preference	Features of the cost model — least-cost debit is what the cost binding does on debit	4
Decline code aware, alternate processor on retry, alternate network on retry, step-up aligned, scheduled retry window, dunning optimised	Behaviour ***inside*** the recovery binding	3
Local acquiring preference, local entity selection, domestic scheme preference, settlement currency aligned, dual brand selection, signature versus PIN, message model selection, token requestor aligned, wallet inherited	**Split** — the legal part is policy, the economic part is a feature of the cost and approval models	2 and 4
Volume commitment, tiered pricing aware, settlement speed priority, merchant elected, platform directed, franchise standard	The directed binding, with a disclosure requirement attached	3

THE CATALOGUE IS STILL WORTH KEEPING

As a checklist of considerations, not a deployment plan. When a new vertical or market arrives, it is the fastest way to ask what the route has to be able to do — which is a layer-one question, and answering it adds a predicate rather than a strategy.

Complexity has not disappeared. It has moved from a place where it needed seventy-three owners to places where it needs two or three: a capability matrix owner and a model owner. What remains genuinely hard is that the capability matrix must be **verified against processors, not asserted by them**, and that cost tables must be the same data billing uses or the merchant statement will not reconcile. [A]

Bindings, postures and objectives

BINDING	CLASS	OBJECTIVE	FALLBACK
`Route_MaxApproval`	Learned	Approval-first	`Route_Balanced`
`Route_Balanced`	Learned	Exchange rate	`Route_MinCost`
`Route_MinCost`	Parametric	Cost-first	`Route_StaticPreference`
`Route_Directed`	Deterministic	Honour the direction; report inert where infeasible	`Route_StaticPreference`
`Route_Recovery`	Parametric → learned	Inherited, conditioned on the prior outcome	`Route_StaticPreference`
`Route_StaticPreference`	Deterministic	None — an ordered list, health-filtered	**Terminal**

Segment differentiation comes from the capability filter, the policy set, the guardrails and the autonomy level — none of which contend — not from more strategies.

The four objective forms

FORM	EXPRESSION	REQUIRES FROM THE MERCHANT	USE WHEN
Approval-first	`max E[approval] s.t. cost ≤ ceiling`	A cost ceiling in basis points	Margin per transaction is high, or a lost sale is permanent
Cost-first	`min E[cost] s.t. approval ≥ floor`	An approval floor	Thin margin, high volume, small ticket
Economic	`max E[approval × margin] - E[cost]`	Gross margin per transaction	Only where the merchant will disclose margin — rare
Exchange rate — the default product contract	Accept up to ***N*** basis points of additional cost per 100 basis points of expected approval improvement	**One number**	Everywhere else. The practical form

WHY THE EXCHANGE RATE IS THE ONE THAT WORKS

The economic form is blocked on data we do not have: merchant margin is often unavailable and always variable. The exchange rate sidesteps it entirely — the merchant states a basis-point tolerance rather than disclosing their economics.

Implied values, from how a merchant answers: “always take the sale” is ***N*** around 60 or more; “worth it for a meaningful gain” is 15 to 25; “only if it is nearly free” is 3 to 8; “never pay more” is zero, which resolves to cost-first. ***N*** is bounded above by an entitlement ceiling so a merchant cannot breach a concentration or risk constraint by implication.

This contract has been proposed, adopted as the default in one document, and still appears as an unowned open question in another. It is open item D3, and it has been open for four rounds with an answer sitting on the table.

Posture — the one field most merchants ever touch

	MAXIMISE_APPROVAL	BALANCED	MINIMISE_COST	DIRECTED
Objective	max E[approval]	max E[approval × margin] − E[cost]	min E[cost]	Honour the direction
Subject to	cost ≤ ceiling	—	approval ≥ floor	the direction is feasible
Default bound	+25 bps versus the cheapest feasible	—	−15 bps versus the best feasible	—
Autonomy ceiling	3	3	3	**1**
Exploration	Permitted, budgeted	Permitted, budgeted	Permitted, narrow	**None**
Right when	High margin per transaction, or a declined customer is lost permanently	Neither dominates, or the merchant has no view	Thin margin, high volume, small ticket, the customer returns	A contractual or operational reason to fix the route
Wrong when	Margins thin enough that basis points outweigh a recovered sale	*Rarely wrong — it is the safe default*	The decline is final and the customer is lost, so the saving is illusory	Used as a substitute for trust. Usually costs approval

HOW TO ELICIT IT, WITH THEIR OWN NUMBERS

“A transaction at your median ticket is declined that would have been approved on a route costing twelve basis points more. Did we make the right call?”

“No, take the sale” is maximise approval. “It depends on the amount” is balanced, and then elicit ***N***. “Yes, we cannot pay that” is minimise cost. **Asked in the abstract, every merchant says maximise approval**, which is why the question has to carry a number and the number has to be theirs.

Review triggers: ticket size shifts by more than 30%, the margin structure changes, the merchant disputes a decline pattern, the annual review, or a profile change.

The candidate is a tuple, not a processor

```
candidate = { credential_form, network, acquirer, processor,
              message_model, authentication_path }

plus  economics    estimated_cost_bps · settlement_currency · fx_margin_bps · local_acquiring
      performance  predicted_approval (with interval) · latency p50/p99 · availability_state
      constraints  risk_constraints[] · concentration_impact · capability_gaps[]
      provenance   which platform supplied what, and as_of per field
```

The set is bounded before scoring rather than being an unbounded cross product. A candidate with any capability gap is **excluded, not scored with a penalty**. Every exclusion is recorded with a reason code. A field whose source is unavailable is marked stale and the strategy's stale-attribute behaviour applies — it does not silently default.

Routing returns a **ranked list, not one route**, with a TTL measured in seconds. A connection failure uses rank two immediately with no re-decision, because the decision is still fresh and re-deriving it costs latency during an incident. An issuer decline is different: it is new evidence, and it re-enters attempt strategy.

07

Fourteen segment profiles

> Fourteen profiles resolving to five bindings gives the same segment differentiation as fourteen strategies, with a fifth of the ownership, review and contention surface.

#	PROFILE	POSTURE → BINDING	AUTONOMY	THE DISTINGUISHING REQUIREMENT
1	SMB managed	Balanced	3	Pooled cohort model — per-merchant volume never clears the evidence test
2	Mid-market balanced	Balanced	3	Per-merchant model where volume allows
3	Enterprise domestic	Maximise approval	**2**	Advisory: the model recommends, humans approve
4	Enterprise global	Maximise approval, directed in contracted markets	2, per market	Local acquiring entities, multi-currency settlement, residency compliance
5	Subscription recovery	Maximise approval; retries via recovery	3, **joint scoring on**	MIT indicators, network transaction id continuity, account updater
6	B2B invoice	Minimise cost	3 on rail and routing	Level 2 and 3 submission, multi-rail, large ticket. **Rail selection is the high-value decision here**
7	Marketplace	Balanced	3, platform sets for the estate	Split settlement, submerchant obligations, payout and reserve handling
8	Platform / PayFac	Directed, bounded by the delegation grant	**Per grant**	Submerchant hierarchy, delegated authority
9	Franchise	Directed, store exceptions where the agreement permits	1–2	Store and terminal hierarchy, card present
10	Omnichannel retail	Balanced	3	Payment identity continuity across channels — and note PD-001 has no PAR, which is what that continuity actually depends on
11	QSR / unattended	Minimise cost	3, **latency as a hard constraint**	Low-ticket qualification, offline and floor limits, deferred authorization
12	Travel / lodging	Maximise approval	**2** — dispute exposure warrants review	Incremental authorization, delayed presentment, itinerary and folio data
13	Digital goods	Balanced, fraud-weighted	3, fraud guardrails tightened	Card-testing controls, velocity, device intelligence
14	Agentic commerce	Directed **until evidence exists**	**1** — deterministic only	Scoped credential, mandate verification, spend limits, no raw PAN exposure

What a merchant may change

FIELD	WHO SETS IT	MERCHANT MAY CHANGE
Profile	Onboarding, from vertical and volume	By request, with review
Posture	Profile default	**Yes — this is the main lever they own**
Autonomy level	Profile default	Downward freely; upward with agreement
Processor exclusions	Merchant	Yes — expressed as ***policy***, not preference
Rail options offered	Entitlement	No — commercial, not configuration
Capability requirements	Derived from vertical	No — they are facts about the transaction
Policy set	Compliance, network, PSP	No, and their own policies cannot override upward
Guardrail thresholds	Platform	Tighter yes, looser no
Experiment participation	Platform	Opt out, honoured at assignment

Anything a merchant sets that cannot take effect is **reported inert, with the reason and the authority level that blocked it**. Silently ignoring it is the failure mode.

The evidence test, applied per population

```
transactions per month in this population   ≥ 50,000
distinct issuers observed                   ≥ 200
routes with non-trivial observed volume     ≥ 2
months of propensity-logged history         ≥ 3
exploration budget available                > 0

all five  ⇒  a learned binding is available
any one missing ⇒ pool into a broader cohort, or stay parametric
```

Do not lower the bar. A learned strategy on thin evidence is worse than a deterministic one, because it is confident. Pooling is the answer: 1,400 merchants at a median of 900 transactions a month each individually fail, and pooled by vertical, region and ticket band they reach 210,000 a month and pass. [C]

Recovery is not ordinary routing

> The reflex error: routing a retry to a different processor by default. On a soft decline that authentication would clear, changing processor discards the issuer relationship that was about to approve.

RECOVERY MUST RESPECT	WHY
Decline class	Soft, hard, technical or unknown determines which actions exist at all
Retryability	A hard decline is never retried, whatever the economics look like
Idempotency	The retry carries the same key family. A duplicate authorization is a worse outcome than a decline
Authentication state	If authentication completed, the retry carries the evidence forward. It does not re-authenticate
Credential state	An updated credential changes the candidate set; a suspended one removes it
Scheme limits	Excessive-retry rules bound attempts per period independently of our budget
Timing	Some declines require delay, not an alternative. ***Where*** is a second question to ***when***
Status uncertainty	An unknown outcome is not a decline. Recovery owns it; routing does not retry it

The boundary with attempt strategy: attempt strategy decides **whether** to try again, what kind of action, and against which budget. Recovery routing decides **where**, once that is settled. Two decision points, two bindings, two attributions — and neither may take the other's decision.

THE TERMINAL FALLBACK, AND THE CORRECT FAILURE

The chain is `MaxApproval → Balanced → MinCost → StaticPreference`. The terminal is an ordered processor list from merchant configuration, health-filtered, with no model, no remote lookup and no fallback of its own. It must be exercisable with the control plane unreachable, and it is **exercised monthly as a scheduled test** — an unexercised fallback is a plan, not a capability.

If the terminal cannot produce a decision, the transaction fails with `NO_FEASIBLE_ROUTE`, and **that is the correct outcome**. An improvised decision is worse than a declined one, because it is unattributable and unrepeatable.

The policy platform

> Policy determines what is permitted. Decisioning determines what is optimal. Orchestration executes the result.

SEVEN VERBS · AND HOW THEY COMPOSE DOWN THE AUTHORITY LADDER

ALLOW	**permits what would otherwise be unavailable**	additive within a level; cannot restore a higher DENY
DENY	**removes a capability for this context**	absorbing — no lower level restores it
PROHIBIT	**removes one action within a capability**	absorbing
REQUIRE	**narrows to the required subset**	intersecting; multiple REQUIREs intersect
BOUND	**constrains a numeric or enumerated range**	intersecting; the tightest bound wins
PREFER	**orders whatever remains**	never adds or removes; a lower level may reorder
DELEGATE	**transfers bounded authority downward**	the only way authority moves down, and it expires

REGULATOR → NETWORK → PSP → PLATFORM → PAYFAC → MERCHANT → SUBMERCHANT → LEARNED STRATEGY

Optimization sits at level eight, below every human and regulatory authority.
It chooses among options levels one to seven have already declared acceptable. It never expands the set.

— Numeric priority orders policies within a level only. A policy declaring a priority intended to cross a level fails compilation, which is how a whole class of privilege-escalation error is removed rather than monitored.

Namespace is identity, not a folder

The namespace determines the authority level, the owner, the approval chain and the precedence behaviour. **Authority is derived from the namespace, not declared**, which removes a whole class of privilege-escalation error rather than monitoring for it.

```
policy/
  regulatory/{us|eu|uk|apac}/{country}/   REGULATOR   compliance
  network/{scheme}/                       NETWORK     network relations
  psp/{trust|safety|appetite}/            PSP         institutional risk
  risk/{fraud|card_testing|ato}/          PSP         risk
  product/{authentication|credential|routing|...}/  PSP   product
  platform/{id}/   isv/{id}/              PLATFORM    the platform or ISV
  payfac/{id}/     franchise/{id}/        PAYFAC*     the PayFac or franchisor
  merchant/{id}/                          MERCHANT    the merchant
  store/{id}/                             SUBMERCHANT merchant or franchisee
  operations/{incident|maintenance|emergency}/  PSP   operations

* franchise authority is configurable per agreement and recorded on the relationship,
  because a franchisor's power over a franchisee varies by contract and jurisdiction
```

RISK IS NOT ITS OWN RUNG

Risk predictions are **context attributes** that any policy may reference; risk policies sit at PSP authority, alongside institutional risk appetite. A proposal to make risk a precedence level of its own was declined, because it hides the ambiguity rather than resolving it.

The ambiguity being removed: does a risk rule outrank a network mandate? **It does not.**

Compilation, and what it catches before deployment

CONFLICT CLASS	DETECTION	RESOLUTION
Direct contradiction	Two policies, same capability, same scope, opposing verbs	Higher authority wins; the lower is reported inert; the author is warned at compile time
Unreachable policy	Conditions that can never be satisfied given higher-level denials	Compiles with a warning, reported as inert
Scope overlap at the same level	Intersecting scope, different effects	Priority orders them. **Equal priorities fail compilation**
Bound contradiction	Two bounds producing an empty range	**Fails**, naming both policies
Delegation breach	A delegated policy exceeding its grant	**Fails**
Illegal exception	An exception against a regulator or network policy	**Fails**
Vocabulary drift	An unknown or deprecated attribute	**Fails**, or warns inside the migration window

The attribute vocabulary is governed, versioned and **closed**. An attribute not in it fails compilation rather than evaluating to null at runtime — which is how `fraudScore`, `fraud_score` and `risk.score` come to coexist and mean different things. Renaming is a deprecation with a migration window, never in place, and every attribute has exactly one source domain.

Exceptions

First-class objects with a target, a scope always narrower than the policy they except, a mandatory free-text reason, an approval chain, an authority level that cannot exceed the policy, a review date, and **a mandatory expiry — an exception without one does not compile**. Expired exceptions lapse silently and the policy reasserts. Exception count and age are reported to product monthly, because exceptions are the sprawl vector and they are owned by product, not by whoever requested them.

An exception may never except a regulator or network policy. Those require a policy change with the same approval chain.

THE RUNTIME HAS NO ENDPOINT

A hot-path policy evaluation API was proposed twice and declined twice. It contradicts the same documents' own correct conclusion that the runtime must be local and bundle-driven, and **publishing the endpoint guarantees someone calls it from the payment path.**

Runtime evaluation is an in-process library call over the held bundle: no network call, no database query, no cache-miss path. Five API families exist and none of them is on the payment path — management, check ("can I?"), simulate, explain and deploy. [A]

Four policy shapes

Nearly every policy anyone actually writes is one of four shapes. Worth having the four right.

SHAPE	THE MECHANIC THAT MATTERS
Block a specific BIN	A membership test over an enumerated list. The easy case.
Block a BIN range	**Ranges are expressed as full-length account-number intervals, not six or eight digit prefixes.** Scheme range files are published as intervals, prefix lengths vary by network, and they have changed over time.
Restrict a currency within a geography	Two attributes, both from context assembly. Straightforward once the vocabulary is closed.
Block a geography	The one that is four rules wearing one name — see below.

"BLOCK COUNTRY X" IS FOUR DIFFERENT RULES

Issuer country (where the card was issued). Billing country (what the cardholder typed). IP country (where they appear to be). Merchant country (where we acquire).

A US-issued card used by a traveller in Europe against a UK merchant produces three different answers. So **the policy language has no bare `country` attribute**, deliberately. An author must name which one they mean, and a policy that tries to mean all four has to say so explicitly.

BIN matching mechanics

The BIN and its range arrive with the credential, resolved at vault time. **There is no runtime BIN lookup** — that is what keeps constraint evaluation inside three milliseconds. Ranges compile into an interval tree keyed by numeric account range: a logarithmic lookup over a few hundred thousand intervals, in memory, with no allocation on the hot path. The compiler merges overlapping intervals, detects containment, and reports a range wholly contained in another as inert.

ATTRIBUTE STATE	BLOCKING POLICY (deny at regulator or PSP authority)	OPTIMISING DECISION
Fresh	Evaluate normally	Evaluate normally
Stale	Evaluate normally, flag	**Suppress optimisation**, fall back to the configured default
Missing	**DENY**, with `ISSUER_ATTRIBUTES_UNAVAILABLE`	Suppress optimisation

Blocking policies fail closed; optimising decisions fail open. Both are deliberate, and they look like an inconsistency until you notice they are protecting different things.

What happens to stored credentials when a block lands

New authorizations are denied from the moment the bundle is in force. Matching stored credentials are marked **policy-suspended, not deleted**; network tokens are suspended, not deleted, because deletion forfeits the token and the provisioning relationship. Account updater is suspended. Recurring mandates are flagged and the merchant is notified with the affected customer count, not the identities, unless they are entitled to them. In-flight authorizations complete under the constraint set bound at their start; captures against prior authorizations are permitted; **refunds are always permitted — a block must never trap a cardholder's money**; and disputes are unaffected.

PART

III — Evidence

11 What gets recorded

12 Registration, promotion and the ratchet

13 Attribution and measurement

14 Guardrails

15 The worked example

What gets recorded

> Three record types, and two fields inside them that are the difference between a platform that can learn and one that cannot.

RECORD	ONE PER	CONTENTS
Execution record	Payment attempt	Attempt and intent ids · transaction type · the flow that ran and its version · experiment, arm and assignment ids · strategies applied with versions · artefact versions (constraint set, BIN snapshot, routing table, models) · constraint provenance — which authority produced which bound · timings
Decision record	Decision point per attempt	Strategy and version · `action_space[]` (what was permitted) · `considered[]` · `action_taken` · `rejected[]` with reason codes — the "why not" view · `fallback_would_have_been` · `regime` · `propensity` · `duration_ms`
Outcome	Attempt, joined several times	Approval and auth code · decline class · cost (interchange, scheme, processor) · fraud flag and amount · dispute flag and amount · which horizons have closed

NEITHER PROPENSITY NOR THE FALLBACK CAN BE BACKFILLED

Without propensity there is no off-policy evaluation, and the platform can only learn from live experiments — which means learning at the speed of calendar time rather than at the speed of traffic. Without the fallback counterfactual there is nothing to attribute against.

Neither can be reconstructed for traffic that has already run. **Logging them should begin on the current engine before this platform exists**, with no behaviour change, because it is cheap to log and impossible to retrofit. It is the only item in the whole programme where a month of delay is a month of permanent loss.

Propensity is 1.0 for deterministic strategies, and it is logged anyway. [A]

The absent-block rule

A transaction that never reached the assignment point has **no experiment block and no eligibility block** — absent, not `eligible: false`. Recording an eligibility verdict it never underwent would inflate every denominator in the readout by the size of the population that never got there, which in the worked example is about 62%.

Outcome horizons

MEASURE	HORIZON	WHAT IT IS USED FOR
Approval	Seconds	The primary metric, in most experiments
Cost	2 days	Cost of acceptance guardrail
Fraud	30 days	Guardrail at the proxy, evidence at the confirmed measure
Dispute	120 days	**Full ramp waits on this one**, even after promotion at 50%

Registration, promotion and the ratchet

> Thirteen registration gates, every one mechanical. None requires judgement, which is the property that makes them survive a deadline.

GATE	APPLIES TO	REJECTS
Population uniqueness	Strategy	A population intersecting another active binding at the same decision point
Action-space binding	Strategy	A strategy that could select an action outside the permitted set
Contract completeness	Strategy	Inability to emit propensity, the fallback counterfactual and the rejected list
Fallback named	Strategy	An implicit or absent fallback, or a chain not terminating
Naming	Strategy, flow	Sequential or non-conforming names
Expiry present	All three	Anything without a review date and an expiry
Single difference	Flow	`diff_count ≠ 1` on a variant
Experimentability	Flow	A variant differing at a non-experimentable decision point
Binding completeness	Flow	Any unbound decision point
Layer and segment uniqueness	Experiment	A second active experiment on an occupied layer and segment
Allocation	Experiment	Arms summing above 1.0 less the holdout
Unit of randomisation	Experiment	A unit below the level at which the primary metric is measured
Pre-registration	Experiment	Missing hypothesis, primary metric, sample size or stopping rule

THREE PROMOTION GATES · NONE SKIPPABLE, EACH CATCHING WHAT THE OTHERS MISS

GATE 1 — LOG ADEQUACY

effective sample size sufficient in every segment where the candidate acts differently; overlap confirmed; clipping mass below threshold

catches an estimate computed where the logs hold no evidence

GATE 2 — OFF-POLICY ESTIMATE

doubly robust estimate with an interval excluding harm, per objective term, per affected segment

catches a candidate better on average and worse for an identifiable population

GATE 3 — LIVE ARM

directionally consistent with the estimate, within the interval, at the pre-registered sample size, fidelity above floor

catches systematic estimator bias no held-out data reveals

VARIANCE ALONE IS THE WRONG CRITERION

Variance falls when the behaviour policy was nearly deterministic — exactly when the logs contain no evidence about the alternatives.

— A candidate can fail gate one with a beautifully tight variance number. On promotion the baseline ratchets: the strategy becomes the default binding, a new baseline flow is compiled, and the next experiment measures against that.

THE RATCHET

On promotion the strategy becomes the default binding for its population, a new baseline flow version is compiled with it bound, the experiment closes, and the next experiment on that transaction type measures against the new baseline.

This is what stops the platform drifting. A comparison against a binding from two quarters ago is not a comparison against current — and background improvement accrues to the baseline too, which is why the comparison must be concurrent and never against a historical period.

Everything expires. A flow at zero exposure for ninety days is archived. An inert strategy is reported as inert. An expired experiment closes as inconclusive and is logged as such, because it consumed traffic and produced nothing and that should be visible. An unreviewed strategy is suspended rather than silently continued. Rollback is a pointer change.

Attribution and measurement

> Every decision logs what it did, what its declared fallback would have done, and the propensity. Contribution is the estimated difference against that fallback, over the strategy's own population, in the objective's units, with a confidence interval — always.

Three honest limits, stated in every readout

LIMIT	WHAT THE READOUT SAYS INSTEAD
Overlapping strategies on one transaction cannot be attributed independently	Reports **joint contribution** and names the overlap, rather than dividing it by assumption
Attribution is against the declared fallback, not a world with no strategy at all	Says so. The second is not estimable and claiming it would be dishonest
A decomposition across decisions not all randomised in the same period is not identifiable	Reports main effects for what ***was*** randomised, plus an explicitly **unattributed remainder**

WHY THE UNATTRIBUTED REMAINDER IS PRINTED RATHER THAN APPORTIONED

A decomposition of the form "18 basis points from routing, 11 from credential selection, 9 from authentication, 4 interaction" is exactly what leadership wants, and it is identifiable only where every one of those decisions was randomised in the same period.

Where they were not, apportioning the remainder by plausibility produces numbers the organisation plans against — which is worse than saying we do not know. This creates a real tension with any scheme that rewards teams on attributed basis points, and that tension is unresolved. [J]

Treatment fidelity

The share of transactions assigned to an arm that actually received its treatment. A transaction may not receive it because the decision point fell back or breached its deadline.

Both naive responses are wrong. **Do not drop it** — that biases toward periods when the platform was healthy. **Do not ignore it** — an arm at 96% fidelity and one at 99.9% are not the same experiment. The transaction stays in its assigned arm, the fallback is flagged, and fidelity is reported alongside the effect.

THE FLOOR, WHICH TWO SOURCE DOCUMENTS STATE DIFFERENTLY [CONFLICT C5]

Reading A: below 95%, ***or three points below control***, blocks promotion.
Reading B: below 95%, ***or a gap of more than two points between arms***, blocks promotion and triggers investigation.

These are not the same threshold, and they diverge exactly where it matters — in a low-volume arm where fidelity is noisy. **Open. Owner: data science.** Until it is settled, use the stricter of the two, which is B.

Whichever is chosen: analysis groups by ***arm assigned***, always — the intention-to-treat population — and fidelity is reported separately. Grouping by what was actually used silently moves fallen-back transactions into the control arm and biases the estimate toward zero.

Interaction, calibration and stopping

	RULE
Interaction	Orthogonal assignment does not mean orthogonal effects. An authentication arm changes the decline mix a retry arm then operates on. Every readout tests interaction with concurrently live layers, and **a significant interaction blocks promotion until both are analysed together**.
Calibration	A challenger runs alongside the champion, decides nothing, and records what it would have chosen and what it predicted. Calibration error is measured on the decisions where they ***agree***. A challenger whose predictions are wrong on the cases we can check should not be promoted on the strength of the cases we cannot.
Stopping	Run to the pre-registered sample size and read out once. **No peeking at the primary metric.** Continuous monitoring must be declared at registration and uses a sequential design with alpha spending. Ad-hoc daily checks against a fixed-sample design inflate false positives to the point where the result carries no information — and in this domain the reader is usually motivated to stop on a good day.
Guardrails	Exempt from the no-peeking rule. They are monitored continuously by design, because their job is to stop harm rather than to measure an effect.

Guardrails

Defaults. A profile may tighten any of them; none may be loosened without the approval recorded against the profile.

GUARDRAIL	THRESHOLD	ACTION
Primary metric	Arm below control beyond the interval, two consecutive windows at ≥ 2,000 per arm	Abort the arm
First-attempt approval	Below baseline while transaction approval holds	Halt ramp — recovery is masking a worse first decision
Per-merchant harm	Any merchant below control beyond the interval at ≥ 5,000 transactions	Abort the arm, **even if the aggregate is positive**
Fraud proxy	+15% relative — ***or*** reported fraud above 1.5× control at ≥ 5,000 per arm **[conflict C6]**	Abort the arm
Dispute proxy	+10% relative — ***or*** fraud-coded dispute rate above 1.3× control **[C6]**	Halt ramp, review at the confirmed horizon
Cost of acceptance	+10 bps versus control, or above the posture's ceiling	Abort the arm
Latency	p99 above budget by more than 20 ms, two consecutive windows	Halt; abort if sustained
Unknown-outcome rate	Above the recovery SLO	Abort immediately
Treatment fidelity	Below 95%, plus the disputed second clause **[C5]**	Block promotion
Exploration budget	Consumed	Exploit only — no abort
Processor concentration	35% of merchant volume, or 50% platform-wide, rolling 24 hours	**A constraint, not a guardrail** — it bounds the action space before scoring
Constraint violation	Any	Abort immediately, no threshold

THREE THINGS THAT ARE CONSTRAINTS RATHER THAN REWARD TERMS

Latency and reliability. A strategy cannot trade a deadline for approval, however favourable the exchange looks.

Concentration. A guardrail is monitored and tunable by the strategy owner; a constraint bounds the candidate set before scoring. An optimiser able to trade concentration against approval will concentrate, and it will be locally correct every single time it does.

Fraud, at the constraint level. Letting routing trade fraud against approval hands a risk decision to an optimiser that does not own it.

And the fourth thing that is neither: **aggregate improvement is not sufficient.** A strategy raising portfolio approval while lowering it for one large merchant has redistributed, not improved.

Disclosure

Queryable — a merchant may ask which experiments their traffic is in, and see the arm. Opt-out is honoured **at assignment, not filtered at analysis**. There is no per-experiment notification: routine optimisation inside a contracted posture does not generate a notice per cell. Harm detection is not optional. Anything touching amount, currency or a disclosure flow is never an experiment at all.

The risk, stated plainly: if the largest merchants opt out, learning happens on the tail where our evidence is thinnest. **The mitigation is to make participation worth having** — attribution reported to the merchant in their own units — not to make opting out harder. [J]

The worked example

> One experiment, run through registration before launch. It produced seven findings; four blocked registration and four remain open. It is reproduced because it is the best single illustration of what the gates are for.

#	FINDING	SEVERITY	RESOLUTION
1	The variant changes two decision points at once, so a win cannot be attributed to either	Blocking	Split by layer: two variants differing at one point each, two experiments on two layers with independent salts, run concurrently. Costs one extra registration and no calendar time. Three arms and a 2×2 factorial were the alternatives considered
2	Selection criteria do not require the treatment credential to exist, so part of the test arm cannot be treated	Blocking	Predicate corrected to require the treatment credential, active, and the randomisation unit to be resolvable. “Present” was doing the work of “usable”
3	No unit of randomisation declared, and the obvious choice is wrong	Blocking	The customer, not the credential and not the transaction. Guest checkouts are excluded rather than assigned on a fallback key
4	A 95/5 split needs 3.0M eligible transactions to detect 25 bps — 5.3× what an even split needs	Blocking on duration	Two-phase ramp: a 95/5 soak for guardrails, excluded from the primary analysis, then 90/10 to the pre-registered sample size
5	No primary metric, guardrails, sample size, stopping rule or expiry	Advisory	All supplied at registration
6	Assignment placed after the multiplicity gate — correct, and it corrects the PRD	Advisory	**Still open.** The PRD amendment blocks full ramp. Conflict C1
7	The two sub-decision-point identifiers are labelled inconsistently between two sets of notes	Advisory	**Still open, and blocking registration.** They land in strategy names, layers, decision records and every readout. Conflict C8

THE SPLIT ARITHMETIC, WHICH IS THE GENERALISABLE PART [C]

At 80% power, 5% two-sided, against an 87% baseline, to detect 25 basis points:

```
95/5    149,510 variant  +  2,840,697 control  =  2,990,207     5.3×
90/10   157,816 variant  +  1,420,348 control  =  1,578,165     2.8×
80/20   177,544 variant  +    710,174 control  =    887,718     1.6×
50/50   284,070 variant  +    284,070 control  =    568,139     1.0×
```

Moving from 95/5 to 50/50 *increases* what the test arm needs, from 150k to 284k, while *halving* total traffic from 3.0M to 568k. An imbalanced split minimises exposure to the variant, which is exactly what you want while the guardrails are unproven — and it maximises total traffic consumed, which is exactly what you do not want once they are. Hence the two-phase ramp.

Pooling a 95/5 period with a 90/10 period without time stratification produces a sample-ratio mismatch that most analysis tooling will flag and some will silently absorb.

The readout, at day 37

MEASURE	CONTROL	VARIANT	VERDICT
Eligible, phase 2 only	1,420,200	157,800	Phase 1 soak of 364,200 excluded from the primary analysis
Treatment fidelity	99.94%	97.22%	Pass against a 95% floor — 2.78% fell back
Transaction approval rate	**87.31%**	**87.68%**	**+37 bps, CI [+20, +54]**. Intention-to-treat. Fidelity-adjusted the implied treated effect is +38 bps; the readout reports the ITT figure
Fraud proxy at 30 days	—	+0.4% relative, CI [−1.9, +2.7]	Within
Dispute proxy	—	+2.1% relative	Within; the confirmed 120-day horizon is still pending
Cost of acceptance	—	+2.1 bps	Within — but see below
Interaction with the concurrent layer	—	+3 bps, CI [−14, +20]	Not significant; both promotions may proceed
Calibration	Predicted +34	Observed +36	Error 2 bps

Decision: promote to 50% exposure. Full ramp and the baseline recompile are held until the 120-day dispute horizon closes.

TWO THINGS IN THAT READOUT THAT DESERVE ARGUMENT, NOT ACCEPTANCE

Cost rose 2.1 basis points and the guardrail passed anyway. That is a trade, not a free win. If this merchant's posture were minimise cost, the same result reads differently and might not promote at all. A readout that reports a pass without naming the trade has under-reported.

A secondary guardrail rose 11% against a 20% threshold, and 37 days is not long enough to know whether it plateaus. Promotion to 50% doubles the pressure on it. An experiment that improves approval while quietly degrading an estate the platform will want later has borrowed from a capability rather than created one.

Separately, and not an arm of the experiment: the permanent holdout read 86.94% against a full population of 87.55% — a cumulative **+61 bps**, which is the aggregate effect of every strategy promoted at every decision point since the holdout was reserved. That number is the reason the holdout exists, and it cannot be produced any other way. [C]

PART

IV — Vocabularies and readiness

16 Vocabularies
17 Autonomy
18 Glossary
19 Readiness

Vocabularies

> Eight vocabularies, and one of them has to be fixed before anything else is worth measuring.

Approval rate — fix this one first

METRIC	NUMERATOR	DENOMINATOR	USED FOR
Attempt approval rate	Attempts approved	Attempts dispatched	Processor and route performance. **Never** for comparing strategies that differ in retry behaviour
First-attempt approval rate	Payments approved on attempt 1	Payments with ≥ 1 attempt	Routing, credential, authentication, message strategies — anything not changing retry behaviour
Payment approval rate	Payments with any approved attempt	Payments initiated	Attempt strategy, recovery, dunning. The only honest denominator when retry behaviour is what changed
Recovery rate	Payments approved on attempt ≥ 2	Payments whose first attempt was declined	Recovery strategies, alongside payment approval rate
Fraud-adjusted approval rate	Approved payments less confirmed-fraud payments	Payments initiated	Any strategy whose guardrails include fraud

The selection rule: does the strategy change retry behaviour? If not, first-attempt approval rate. If so, payment approval rate, and report attempt count alongside it. ***A readout quoting an approval rate without naming which one is not a readout.*** Every denominator excludes screening blocks, policy blocks, validation failures and idempotency-suppressed duplicates.

Canonical decline taxonomy

CODE	CLASS	RETRYABLE	PERMITTED ACTIONS
`INSUFFICIENT_FUNDS`	Soft	Yes, scheduled	Scheduled retry only. An immediate retry is waste
`DO_NOT_HONOR`	Soft	Yes	Step up, processor substitution, credential substitution
`ISSUER_UNAVAILABLE`	Technical	Yes, immediate	Processor or network substitution
`AUTHENTICATION_REQUIRED`	Soft	Yes	Step up. **Substitution alone will not clear it**
`SUSPECTED_FRAUD`	Hard	No	None. Retrying is a scheme violation risk
`STOLEN_OR_LOST` · `STOP_PAYMENT`	Hard	No	None
`INVALID_CARD`	Hard	No	Credential substitution only, if another exists
`EXPIRED_CARD`	Hard	Conditional	Account updater, then retry on the updated credential
`CVV_MISMATCH` · `AVS_MISMATCH`	Hard	No	None. Merchant action required
`EXCEEDS_LIMIT`	Soft	Yes, scheduled	Scheduled retry, or a partial amount where the merchant permits
`RESTRICTED_CARD`	Hard	No	Credential substitution
`PROCESSOR_UNAVAILABLE`	Technical	Yes, immediate	Processor substitution
`NETWORK_TIMEOUT`	**Unknown**	No, until resolved	**Recovery owns it** — reversal advice, then status inquiry
`DUPLICATE`	Technical	No	None. Retrying a duplicate creates a second one
`INVALID_REQUEST`	Technical	No	None. Merchant or platform defect

The per-processor mapping is a versioned artefact owned by the processor adapter, not by routing. **Where a mapping is uncertain, the safe class is hard.**

Fraud and dispute — three populations routinely conflated

TERM	AVAILABLE AFTER	USED IN
Reported fraud — asserted, unconfirmed	Days	**Early guardrail proxy only**
Confirmed fraud — issuer or scheme confirmed, or an unrepresented fraud-coded chargeback	Weeks to months	Fraud-adjusted approval, and the objective's fraud term
Fraud-coded dispute — regardless of outcome	Weeks	Scheme monitoring exposure, ***not*** the fraud term
Non-fraud dispute — service, quality, processing error, cancellation	Weeks to months	The dispute term. **The larger population for most merchants, and absent from every "fraud rate"**
Net dispute loss — amount lost after representment, plus fees	Months	The economic objective

Never promote on the proxy and never abort on the confirmed measure — the first is too slow and the second too noisy for the job it would be doing.

Reason codes, and versioning boundaries

Reason codes are a merchant-visible public contract: canonical, enumerated, never free text. Adding one has an owner and a review; **a code is never removed, only deprecated**; and each carries merchant text, integrator text, a merchant action, retryability and the authority that produced it. The inert codes matter most — a preference shadowed by higher authority, a strategy shadowed by another, a direction that is infeasible — and they are visible to whoever set the thing that is inert.

QUESTION	RULE
New version, or new strategy?	A new version if population and objective are unchanged. **A new strategy if either changes** — attribution against the prior version would otherwise compare different things
Does a parameter change bump a version?	Yes. Every artefact that alters a decision is versioned, including a threshold
Does a model retrain bump a version?	Yes on the model artefact. The strategy version bumps only if its binding changes
What closes a day?	**UTC midnight, universally.** Not merchant local time, not the acquirer cut-off — those are separate fields, reconciled against, never the boundary
Which timezone for an experiment window?	UTC. A window that shifts with daylight saving introduces a seasonal artefact into every readout
When does a policy take effect?	At its validity start in UTC, subject to propagation. The gap between that and propagation completion is **measured and reported, not assumed zero**
Which version applies to an in-flight transaction?	The one bound at context assembly. **Versions never change mid-transaction**

Autonomy

Five levels, set per decision point per client: **0 fixed · 1 measured · 2 advisory · 3 bounded auto · 4 auto with veto**.

DECISION POINT	LEVEL 2 MEANS	LEVEL 3 MEANS	CEILING
Rail selection	The model recommends a rail; a human approves the default per merchant	The model selects per transaction within the offered rails	3
Credential selection	The model recommends the default form; a human sets it	The model selects per transaction, exploration permitted	3
Authentication	The model recommends thresholds; a human sets them	The model decides per transaction	**3, with a reduced exploration budget** — an exploratory challenge costs a real cardholder a real interruption
Routing	The model recommends; humans approve ramps	The model selects per transaction, exploration permitted	3
Attempt strategy	The model recommends the schedule; humans set budgets	The model selects action and timing within typed budgets	3

Level 4 is not proposed for any decision point. Level 5 does not exist. The directed posture caps autonomy at 1 by construction.

THE POSTURE-TO-BINDING MAP, ACROSS ALL FIVE DECISION POINTS

Posture selects the routing binding directly, and influences but does not select the other four:

```
                MAXIMISE_APPROVAL    BALANCED         MINIMISE_COST    DIRECTED
Rail            SpeedAware/single    CostAware        CostAware        Directed
Credential      JointWithRoute*      PreferToken      PreferToken      Directed
Authentication  RiskAdaptive         RiskAdaptive     RegulatoryOnly   MerchantDirected
Routing         MaxApproval          Balanced         MinCost          Directed
Attempt         Standard/Dunning     Standard         Minimal          Directed

* where entitled, else PreferToken
```

Posture is not a licence to ignore constraints. It orders preferences within an action space that capability and policy have already bounded.

Glossary

The most-cited part of the source corpus. Where a term is contested, the contested reading is marked.

TERM	DEFINITION
Action space	The set of actions permitted at one decision point for one transaction: capability ∩ entitlement ∩ policy. A strategy selects within it and cannot expand it
Attribution	The estimated contribution of a strategy, against its declared fallback, in the objective's units, with a confidence interval
Autonomy level	How much the platform decides without a human. 0 fixed, 1 measured, 2 advisory, 3 bounded auto, 4 auto with veto
Baseline	The flow bound to the transaction type with role baseline at the time of comparison. **Ratchets on promotion**, so a baseline from two quarters ago is not the current one
Binding	The one strategy attached to one decision point for one population. Exactly one, always
Champion / challenger	The champion is the bound strategy and it executes. The challenger runs alongside, decides nothing, and records what it would have chosen and what it predicted
Class	Deterministic, parametric or learned. **The boundary is fitted parameters, not sophistication**, and it determines the governance burden
Cohort	Merchants pooled for modelling because none individually clears the evidence test. Defined by vertical × region × ticket band unless stated otherwise
Counterfactual calibration	Error between a challenger's prediction and the observed outcome, measured only where champion and challenger agree
`diff_count`	Decision points at which a variant differs from its baseline. Must equal 1 or the flow does not compile
Exploration budget	Basis points of approval rate on the scope's own volume that may be spent learning, per period. **Never a share of transactions.** Platform-level ownership is open item D4
Feasible / eligible / available	Feasible passes capability and policy. Eligible adds commercial entitlement. Available adds live health. Three different words, deliberately
Feature	An input a strategy reads. Not a strategy: no binding, no contention, no attribution question
Holdout	A fixed share of every population that receives baseline flows and is never assigned to any experiment ***or exploration***. 2% default. **Cannot be established retroactively**
Inert	A setting, preference or strategy that can never take effect because something of higher authority precedes it. Reported as inert, never silently ignored
Joint scoring	Evaluating credential and route as a tuple rather than sequentially. A better optimum, shared attribution, and the two decision points then share one experiment layer
Layer	A decision point with its own hash salt, so assignment across layers is independent
Posture	The merchant's stated tradeoff between approval and cost. Selects the routing binding
Profile	A named preset of capability requirements, policy set, posture, autonomy and guardrails. One per merchant per transaction type
Propensity	The probability the chosen action would be chosen, given context and policy version. 1.0 for deterministic strategies, and logged anyway
Regime	What produced a decision: policy (forced), heuristic (rule), learned (model) or fallback (degraded). Recorded on every decision
Segment	A controlled-vocabulary population label the allocation ledger enforces uniqueness against. **Not a marketing segment, and it does not yet exist**
Treatment fidelity	The share of an arm that actually received its treatment. An arm below the floor does not promote, regardless of what it measured
Transaction type	The classification a transaction resolves to, naming the eligible flows. Format: region, channel, funding, mandate
Unit of randomisation	The object hashed to assign an arm. Must be at or above the level the outcome is measured

SEVENTEEN TERMS USED BUT NEVER DEFINED

The source corpus uses these without definitions, and each is currently being assumed differently by at least two teams. Proposed definitions exist; each needs an owner to confirm it:

cost of acceptance (excludes our own margin) · approval delta versus default · per-merchant harm · minimum sample floor · strategic account (a relationship judgement, not a revenue threshold) · amount band · processor health and degraded · credential age (days since first successful authorization, ***not*** since vaulting) · attack state · qualification achieved (the tier applied at clearing, not the tier the route was capable of) · recovered transaction · evidence test population · exploration budget · feasible · baseline · **merchant margin** — blocking · **the controlled segment vocabulary** — blocking, and it does not exist.

Settling these is described in the source as a week of conversations. It is the cheapest unblocking work available. [A]

Readiness

#	MUST EXIST BEFORE	ITEM	OWNER
1	Any authoring	Capability catalogue with stable ids, **verified against processors rather than asserted by them**	Product
2	Any authoring	Attribute vocabulary, closed and versioned. Unknown attributes fail compilation	Architecture
3	Any authoring	Transaction type list, with an owner for adding to it	Product
4	Any authoring	The seventeen definitions, confirmed	Named per row
4a	Any guardrail or readout	The approval-rate denominator named in every metric	Data science
4b	Any merchant-visible decline	Reason code vocabulary published with merchant text	Developer platform
4c	Any attempt strategy	Canonical decline taxonomy plus per-processor mapping tables	Processing
4d	Any strategy	Terminal fallback per decision point, exercisable with the control plane down	Each domain
5	Any policy	Namespace tree with authority mapping and approval chains	Compliance
6	Any policy	Constraint authority owner named	Executive
7	Any strategy	A fallback for every decision point, so nothing is implicit	Each domain
8	Any strategy	Decision contract schema enforcing propensity, fallback and the rejected list	Architecture
9	Any experiment	**Controlled segment vocabulary and the allocation ledger**	Architecture
10	Any experiment	**Permanent holdout reserved. Cannot be created retroactively**	Data science
11	Any learned binding	Propensity and fallback logged for ≥ 3 months. **Start on the legacy engine now**	Routing engineering
12	Any learned binding	Exploration budget owner and ceiling	Product
13	The balanced posture	The margin or exchange-rate decision	Product, finance
14	Any promotion	Counterfactual calibration reporting	Data science
15	Merchant-facing	Inert-setting reporting, so a setting that does nothing says so	Developer platform
16	Merchant-facing	Explanation surface meeting the two-minute budget	Support tooling

THE FOUR BLOCKING GAPS, AND THE THREE THINGS THAT CAN START TODAY

Blocking: merchant margin is not data we have, and the third option — a merchant-set exchange rate in basis points — changes the objective's definition, so it must be settled before the balanced binding is authored. The controlled segment vocabulary does not exist, and without it registration cannot reject overlaps, so the first two concurrent experiments will double-allocate traffic. The exploration budget has no owner, and an unowned budget is spent by whoever asks first. Two resource vocabularies are in circulation and it is cheap to fix now and expensive once it is in an API, a schema, a report and an integration guide.

Can start today, with none of the above: propensity and fallback logging on the legacy engine — no behaviour change, and it cannot be backfilled. The capability matrix, verified against processors — calendar time rather than engineering time. And the seventeen definitions.

Everything else waits on one of those three. [A]

The phases

PHASE	WHAT IT UNLOCKS
0 — Logging	Starts now, on the current engine. Propensity, considered set and fallback emitted from today's decisions, with no behaviour change. **The only item that cannot be caught up later**
1 — Registry	Deterministic strategies deployed without a release
2 — Assignment	The first controlled comparison on a deterministic strategy
3 — Measurement	Promotion on evidence rather than opinion
4 — Learned	Learned strategies, safely
5 — Self-serve	Product authors without engineering. **Should not be promised until the contention model is answered**